

Decision tree: Deriving new objects from a raster

What does the question need?

one raster from several tiles

merge() or mosaic()

both need a shared origin and resolution; merge() keeps the first tile's values in the overlap, mosaic() a statistic

a larger or smaller extent

extend() and trim()

extend() adds rows and columns of NA around a raster; trim() removes outer rows and columns that hold only NA

a sample of locations

spatSample()

method random, regular or stratified; as.points = TRUE for a vector, na.rm = TRUE to skip empty cells

the distance to the nearest feature

rasterize() then distance()

meters in a projected CRS with square cells; a distance surface is a raster, so extract() reads it at points

lines at set values of a surface

as.contour()

levels sets the values; one MULTILINESTRING per level

the connected groups of a class

ifel() then patches()

set every other class to NA first; directions = 8 connects corners; zonal() on cellSize() measures each patch